

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 25

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Waste products that are the raw materials for other processes	1
01.2	• Bacteria fix nitrogen at low temperature using enzymes and energy ultimately from sunlight (renewable) • The Haber process needs high temperature and pressure powered by non-renewable fossil fuels, releasing CO₂	2
01.3	• Incineration uses high temperatures / fuel and releases pollutants; nutrients are not returned as usable raw materials • Decomposition at low temperature recycles nutrients as the raw materials for other natural processes (material cycle)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2-3.6.3.	
02.1	• Less dead organic matter / leaf litter so less humus and slower soil formation; fewer nutrients returned • Loss of canopy and litter exposes soil to rain splash and wind • Decomposing roots no longer bind soil so erosion, gullyng and landslides increase (especially on slopes) • Reduced organic matter and biota lower infiltration and water retention	4
02.2	• Reduced interception and transpiration so less water returned to the atmosphere and less rainfall downwind • Increased runoff: more flooding after rain, lower flows in dry periods; sedimentation of rivers • Increased albedo of cleared land (reflects more sunlight) • Reduced carbon sequestration and loss of the carbon reservoir (CO₂ released by burning / decay) contributing to climate change	4

Question	Marking guidance	Marks
02.3	<i>Any 2 from:</i> • Fragmentation of remaining forest into isolated patches with non-viable populations / edge effects • Loss of inter-species relationships (pollinators, seed dispersers) causing further declines • Loss of genetic resources / potential medicines before they are discovered	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 368-369; AQA spec 3.5.3.3.	
03.1	• Extensive cereal 5.0; intensive cereal 3.0 • Intensive beef 0.1	2
03.2	• Intensive cereal is most productive per hectare (60 GJ) but less efficient (3.0) than extensive (5.0) • Each extra unit of input gives a smaller extra yield (law of diminishing returns) • Beef has very low productivity and efficiency because energy is lost at an extra trophic level	3
03.3	• Any input that increases productivity but requires the use of energy (usually fossil fuel) • Manufacture of nitrate fertiliser (Haber process) and pesticides; fuel for machinery; pumping irrigation water; grain drying; food processing and transport; embodied energy of machinery	3
03.4	• Because of diminishing returns, the first units of input on unfertilised land give the largest yield increases • Moving inputs from intensively farmed land (where extra input adds little) to extensively farmed land gives a greater total yield - though land, cost and distribution limit this	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 324-327; AQA spec 3.5.1.2.	
04.1	• The nucleus of a stem cell from the endangered species is transferred into an empty egg of a closely related species (southern white rhino) • The embryo is implanted into a surrogate female of the related species which gives birth to a northern white rhino • Very low success rates with wildlife so far; deep-frozen cells could allow clones even after the last individuals die	3
04.2	• Micro-propagation: tissue culture producing many cell clusters from one plant, each grown into a genetically identical plant • Black poplar: cuttings / pollarded stems of pure native trees are rooted to produce many identical trees • Needed because the flowers are pollinated by introduced subspecies producing hybrids, so cloning maintains the pure gene pool (but no genetic diversity)	3
04.3	• Stores seeds of wild plants (especially threatened species and crop wild relatives) so species are not lost if extinct in the wild • Seeds collected worldwide; target of 10 000 seeds per species to maintain a diverse gene pool • Stored dry and refrigerated / frozen in an underground reinforced vault to reduce risk of accident • Seeds germinated periodically to check viability and replenish stocks; can be used for reintroduction / breeding	4

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 35-37; AQA spec 3.1.2.3.3.	
05.1	• No major temperature extremes; regular water supply with no pronounced dry season; deep fertile soils built by leaf litter and root binding; strong seasonality • Deciduous canopy present spring to autumn shades the floor during the best growing temperatures • Bluebells, snowdrops, lesser celandine grow and flower early in spring before the trees leaf • Winter food shortage causes animals to hibernate (dormice), store food or migrate	4
05.2	• Long time for species to colonise; some woodland plants colonise very slowly so their presence (ancient woodland indicators) shows the wood is old • Natural woodland has trees of all ages with clearings where old trees died, giving varied light and dead wood habitats • A new plantation has a single-age dense canopy with low light and few niches until trees start to die (over 200 years)	3
05.3	<i>Any 3 from:</i> • Clearance for farmland, plantations, urban development, roads, mineral extraction • Fragmentation isolating remaining woods so populations cannot move between them and are vulnerable • Management change: end of coppicing so woods become uniform and dark; replacement by conifer monocultures • Overgrazing by deer preventing regeneration; introduced pathogens (ash dieback)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-52; AQA spec 3.1.2.3.4.	
06.1	• Surveys monitor photosynthesis rates, biomass, soil water content and the spread of pests / diseases so inputs are applied only where needed • Yield mapping of different parts of a field controls fertiliser application in each area for future crops • This maximises yield while avoiding wastage / reducing leaching and nutrient pollution	3
06.2	• Machinery, pesticides, fertilisers, improved breeds and infrastructure (transport, refrigeration) are expensive • They are mainly available in affluent societies; poorer farmers cannot afford the inputs	2
06.3	• Provides an income that allows producers to afford basic rights such as water, education, health care and food	1
06.4	• Nitrates are leached into water bodies used as sources of potable water • High nitrate levels cause blue baby syndrome (methaemoglobinaemia) and nitrates may be carcinogenic	2
06.5	<i>Any 2 from:</i> • Anaerobic digestion by microbes in rice padi fields • Livestock intestines (ruminant digestion) • Anaerobic decay of manure / slurry	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-332; AQA spec 3.5.1.3-3.5.1.4.	

Question	Marking guidance	Marks
07.1	• Photosynthesis and aerobic respiration balance O₂ and CO₂ in a dynamic equilibrium; evapotranspiration adds water vapour forming clouds and rain; microbes drive the nitrogen and other biogeochemical cycles so wastes do not accumulate and nutrients are recycled • Detritivores, fungi and bacteria decompose dead organic matter releasing nutrients and forming humus that binds soil; organic acids weather rock forming new soil; roots bind soil, reducing erosion and regulating water flow / flooding • Species provide food, pollination (bees, moths, birds, bats - more reliable than wind, especially in forests), seed dispersal (animals carry larger seeds to suitable sites, faeces as fertiliser, gut passage stimulates germination) and habitat provision (trees as nest sites and canopy microclimate; mollusc shells for hermit crabs)	6
07.2	• Its nectar is at the end of a 30 cm tube reached only by the long tongue of the sphinx moth, which carries pollen only between orchids of the same species • Advantage: no pollen wasted on other species and the moth has an exclusive food source; risk: if either species declines the other cannot survive	2
07.3	• They are not ruminants so seeds are not destroyed by repeated chewing / pass through intact; they eat the fruit of most tree species • Not territorial so they carry seeds over large distances / areas	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 18-19; AQA spec 3.1.2.1.3.	
08.1	• Mars has low mass so weak gravity and could not retain a dense atmosphere • Very low atmospheric pressure means water would boil / evaporate away even at low temperatures • Greater distance from the Sun gives low insolation and a mean temperature of -60 °C so water is frozen	3
08.2	• Mass great enough for gravity to retain an atmosphere containing the elements needed (C, H, O, N as CO₂, methane, ammonia, water vapour) and to maintain pressure so liquid water does not boil • Distance from the Sun (the 'Goldilocks zone') gives insolation producing temperatures at which water is liquid (0-35 °C over most of the surface) and proteins are not denatured • Too close (Venus) - too hot; too far (Mars) - too cold	3
08.3	• The molten core produces a magnetosphere that deflects the solar wind / charged particles, preventing biologically damaging radiation reaching the surface • The 24-hour rotation limits heating by day and cooling by night, reducing temperature extremes (Venus' slow rotation gives extreme conditions)	2
08.4	• The tilted axis causes seasonal variation in insolation as the Earth orbits the Sun • Producing seasons (temperature, day length) to which species' life cycles (breeding, migration, dormancy) are adapted; the Moon's gravity keeps the tilt stable and creates tides	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	

Question	Marking guidance	Marks
09.1	• Primary: succession on a newly created habitat with no life / no soil (bare rock, new pond, sand) • Secondary: succession restarting after an existing community is disturbed (tree felling, ploughing, fire, end of grazing) • Faster because soil already exists and many seeds / roots / organisms are still present so there are fewer delays for colonisation	3
09.2	• Human activity (grazing, mowing, burning, coppicing) regularly removes the later successional species so the climatic climax cannot form • The stable community that develops under this repeated disturbance is a plagioclimax • Examples: chalk grassland / hay meadows maintained by grazing or mowing; lowland heath by burning; coppiced woodland; Machair grassland; Broads grazing marshes	3
09.3	• If the traditional activity stops, secondary succession returns the habitat to scrub / woodland and the plagioclimax species are lost • Rainforest and coral reefs are climax communities so conservation aims to minimise human impact	2
09.4	• Extreme environments (deserts, polar) are dominated by abiotic factors; few species tolerate them; populations fluctuate and food webs are simple so changes in one food species have large effects • Stable environments allow many species to evolve inter-species relationships; populations are regulated by biotic factors; complex food webs give stability (rainforests, reefs)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 76-77; AQA spec 3.1.3.3.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Context: wild catches have not increased since about 1990 as stocks are fully or over-exploited; aquaculture output growing rapidly and approaching the wild catch; fish are the last large-scale hunted food • Advantages of aquaculture: control of limiting factors, selective breeding, gender control, high yields per area; fish have low FCRs (no thermoregulation / support) compared with livestock; herbivores (tilapia, carp) and molluscs very efficient; polyculture, IMTA, aquaponics, rice-fish • Limits: carnivorous species (salmon, shrimp) need fishmeal from wild fish so still depend on fishing and deplete forage fish (sandeels - puffins); some species (bluefin tuna) will not breed in captivity; consumer preference for carnivores • Environmental impacts: habitat loss (mangroves), organic pollution, pesticides, antibiotics and resistance, lice, escapes and gene pool contamination, predator control; and how they can be reduced • Fishing can be sustainable with quotas, gear design, NTZs, size limits, effort limits; MSY management needs good data • Judgement: aquaculture can supply much of the growth in demand, especially herbivores and shellfish, but cannot fully replace fishing; both need sustainable management	9
10.2	• A species that is given artificial food (pellets) rather than relying on natural food in the water, eg salmon, shrimp	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-359; AQA spec 3.5.2.	

Question	Marking guidance	Marks
11.1	Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Importance: resources (timber, fuel, food, fibres, medicines) and ecosystem services (atmospheric regulation, carbon store, hydrological regulation, climate, soil conservation, habitat, recreation) • Causes: agriculture (subsistence and cash crops), mineral extraction, reservoirs, urbanisation, transport, unsustainable timber, fuel and fodder exploitation above MSY; trends by region • Consequences: loss of resources, biodiversity and species, fragmentation; hydrological changes; soil impacts; climate impacts (albedo, carbon, rainfall) • Productivity vs biodiversity: non-indigenous species, monocultures, close planting, simple age structure, clear-felling • Sustainable management: MSY harvest rates, mixed indigenous species, mixed age structure, selective logging, traditional coppicing / pollarding, FSC certification, community involvement, protected areas, debt-for-nature, afforestation programmes (China); Rio Bravo case study • Evaluation: economics of slow-growing hardwoods, enforcement, demand from MEDCs, ITTO failures; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-57, 360-369, 386-387; AQA spec 3.5.3, 3.1.2.3.4.	25

Question	Marking guidance	Marks
11.2	Discuss how human activities influence biodiversity, using examples from a range of different contexts. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Direct exploitation: food (fisheries, bushmeat), fashion, entertainment / pets, furniture and ornaments, traditional medicines, other products (whale oil); deliberate eradication of predators, competitors, vectors, pests • Changes in abiotic factors: water availability (drainage, abstraction, HEP), light (forest clearance), oxygen (organic pollution, thermal), nutrients (eutrophication), pH (acid mine drainage, acid rain), temperature (thermal pollution, climate change), turbidity, physical damage; inactivity (stopping plagioclimax management) • Changes in biotic factors: introduced competitors, predators, pathogens, hybridisation; loss of pollinators, seed dispersers, keystone species; food chain impacts (sea otters, puffins, jellyfish) • Habitat destruction: deforestation, farmland expansion, urbanisation, mining, reservoirs; fragmentation and edge effects • Contexts: temperate woodland, rainforest, coral reefs, oceanic islands, mangroves, Antarctica; agriculture, fishing, forestry • Positive influences: habitat creation, conservation management, reintroductions, legislation • Evaluation of which influences are most damaging and hardest to reverse; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26, 328-330, 341-344, 367-369; AQA spec 3.1.2.2, 3.5.1.3, 3.5.2.2, 3.5.3.3.	25